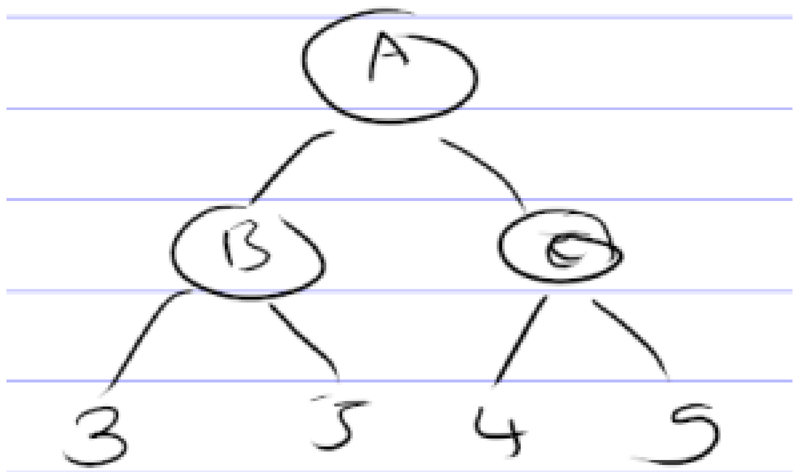
Lab 6

Write a basic program to implement minmax algorithm from game playing theory for the given problem and find the optimal path.

Example problem



Pseudocode

```
BEGIN

 BEGIN
 FUNCTION MINIMAX(depth, node_index, is_maximizing_player, values, max_depth):

   IF depth == max_depth THEN
     RETURN values[node_index] // Return score at leaf node

   IF is_maximizing_player == TRUE THEN
     left_score = MINIMAX(depth + 1, node_index * 2, FALSE, values, max_depth)
     right_score = MINIMAX(depth + 1, node_index * 2 + 1, FALSE, values, max_depth)
     RETURN MAX(left_score, right_score)

   ELSE // Minimizing player's turn
     left_score = MINIMAX(depth + 1, node_index * 2, TRUE, values, max_depth)
     right_score = MINIMAX(depth + 1, node_index * 2 + 1, TRUE, values, max_depth)
     RETURN MIN(left_score, right_score)

 // MAIN
```

DEFINE values = [3, 5, 2, 9]
SET max_depth = 2

CALL MINIMAX(0, 0, TRUE, values, max_depth) → optimal_value

DISPLAY "The optimal value is:", optimal_value
END

<u>Report format</u>

Title: Write a program ……..
Date: Lab assigned date

Objective

- To implement the minmax algorithm from game playing theory and solving the given problem using python.

Theory

- Game playing theory
- Minmax Algorithm

Implementation

- Write down the problem statement
- Python Code
- Output (Screenshots of your output)

Conclusion